

Worksheet - Can Use the Unit Circle (TU)

Angles on the unit circle

For each angle shown, determine the quadrant (I, II, III, or IV) and the cosine and sine of the angle.

If the angle is a multiple of $\frac{\pi}{2}$, enter $+x$, $-x$, $+y$, or $-y$ (as appropriate) for the quadrant.

Angle θ	Quadrant?	$\cos \theta$?	$\sin \theta$?
$\frac{\pi}{3}$			
$\frac{2\pi}{3}$			
$\frac{5\pi}{6}$			
$\frac{11\pi}{6}$			
$\frac{3\pi}{4}$			
π			
$\frac{7\pi}{4}$			
$\frac{11\pi}{3}$			
0			
$\frac{3\pi}{2}$			
$\frac{31\pi}{6}$			
$\frac{11\pi}{4}$			

Rotation notes

By adding $\pm\pi$ or $\pm\pi/2$ to the angle θ (also called “shifting the angle”), the point is *rotated* in the plane by 180° or 90° as follows:

Angle	Rotation
$\theta + \pi/2$	Counterclockwise by 90°
$\theta - \pi/2$	Clockwise by 90°
$\theta + \pi$	Counterclockwise by 180°
$\theta - \pi$	Clockwise by 180°

To determine the effect of rotation upon the *coordinates*, you can either (i) derive it using trig identities as below, or (ii) study similar triangles drawn on the unit circle. Either way, the result is this:

- Angle $\theta + \pi/2$ $\gggg$ $(x, y) \rightarrow (-y, x)$
- Angle $\theta - \pi/2$ $\gggg$ $(x, y) \rightarrow (y, -x)$
- Angle $\theta \pm \pi$ $\gggg$ $(x, y) \rightarrow (-x, -y)$

🔗 Notice that $(-y, x) \rightarrow (y, -x)$ by a 180° rotation!

With trig identities, for example, assuming we start with $(x, y) = (\cos \theta, \sin \theta)$, then for counterclockwise rotation by $\pi/2$:

$$\left(\cos(\theta + \pi/2), \sin(\theta + \pi/2) \right) \gg \gg \left(-\sin \theta, +\cos \theta \right) = (-y, +x)$$

Similarly with the others. You will need to remember the technique from the TD supplemental worksheet, for example:

$$\begin{array}{ll} \cos(\theta - \pi/2) & \gg \gg \cos(-(-\theta + \pi/2)) \\ & \gg \gg \cos(-\theta + \pi/2) \\ & \gg \gg \sin(-\theta) \\ & \gg \gg -\sin \theta \end{array}$$

Rotation practice

Problem 1

Show how to derive *all* these change of coordinate rules:

- Angle $\theta + \pi/2$ $\gg \gg (x, y) \rightarrow (-y, x)$
 - Angle $\theta - \pi/2$ $\gg \gg (x, y) \rightarrow (y, -x)$
 - Angle $\theta \pm \pi$ $\gg \gg (x, y) \rightarrow (-x, -y)$
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Problem 2

If a point with coordinates (a, b) is on the unit circle at rotation θ , then the point on the unit circle at rotation $\theta + \frac{\pi}{2}$ has what coordinates?

Problem 3

If a point with coordinates (a, b) is on the unit circle at rotation θ , then the point on the unit circle at rotation $\theta - \pi$ has what coordinates?

Problem 4

If a point with coordinates (x, y) is on the unit circle at rotation θ , and a point with coordinates (w, t) is on the unit circle at rotation $\theta - \frac{\pi}{2}$, then express w and t in terms of x and y .